

Research on the UAV Intelligent Inspection System to Monitor and Prevent Abnormal Heat of Composite Insulators for 500kV Transmission System

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Abstract—Ensuring a strong and reliable operation of the 500kV transmission system is critical to the energy internet. A large scale of composite insulators in the transmission system were found abnormal heat in a 500kV line after 7 years of operation. Unmanned aerial vehicles (UAV) are powerful tools replacing human power to monitor transmission systems. In this paper, a UAV intelligent inspection system is proposed to monitor and prevent abnormal heat of composite insulators in 500kV transmission system. The reason of the insulators' abnormal heat is well discussed based on infrared images captured by the UAV. By adjusting the position in the shielding rings of running composite insulators, the temperature rise of the composite insulator decreased significantly. The mechanism how the UAV intelligent inspection system operates and the key technology to realize fault monitoring and prevention are well discussed. It is proved that the UAV is promisingly of great feasibility to help the maintenance department to recognize potential fault on the transmission lines and prevent severe faults.

Keywords—energy internet, transmission system, UAV intelligent inspection, abnormal heat, image processing, infrared image

I. INTRODUCTION

As the backbone of the energy Internet, the power grid with huge geographical span and complex energy structure is strongly connected by the transmission system. Ensuring a strong, stable, reliable and economical operation of the transmission system is critical to the energy Internet [1].

Transmission lines, towers and insulators are the basic elements of the transmission system.

It has become a trend to apply UAV intelligent inspection equipment to the transmission system to replace human resources and eliminate hidden dangers of life. From the perspective of the core equipment of transmission system, insulators, new detection technology of composite insulator and fault identification, this paper focuses on how to use UAV technology and communication technology to monitor and prevent abnormal heat of 500kV insulator to ensure the safe and stable operation of the system.

Composite insulators have long been used in power grids since the 1960s. Due to the excellent anti-pollution performance, light weight and easy installation, the application scale of composite insulators is under fast growing. The reliability of composite insulators significantly improves the stable operation of power systems [2]. For now about 7 million composite insulators in China are adopted to the transmission power systems, among which the number of composite insulators used in the southern power grid area has exceeded 860,000.

However, with the operation time growing, the composite insulators frequently experience abnormal defects, for example the abnormal heat may develop to the malfunction of electrical insulation and cause fracture in the composite insulator mandrill. Abnormal heat has caused widespread concern among researchers in our country as well as abroad [3-7].

At present, there are several theories about the occurrence of abnormal heat under operation circumstance. Some studies suggested that partial discharge may lead to

heat generation [8,9]; moisture polarization was summarized in [10-14] as the reason; Some studies also indicated the insulation aging resistance as the reason [15]. Above-mentioned theories can explain the abnormal heat problem of composite insulators to a certain extent. However, there are few studies analyze the operating insulators' abnormal heat by measuring the actual heat composite insulators.

Therefore, it is necessary to carry out research on the problem how to measure the abnormal heat and judge the malfunction online to rapidly response to the hidden faults.

The abnormal heating of composite insulators occurs in a large area of a 500kV line, and the inverters are used for infrared and visible shooting of the insulators of the field lines. In this paper, UAVs(also called as drones) were used to capture infrared images of the corresponding abnormal heat insulators, and upload the corresponding image in time to determine the corresponding fault and implement the processing.

Two composite insulator samples were taken from the operation for test and analysis. The effects of different configurations of the equalizing ring and different fouling conditions on the high pressure end on abnormal heating were studied. It was found that the end field strength distortion and the pollution polarization loss were the key factors for abnormal heating. The control measures for controlling the end field strength distortion are proposed in a targeted manner. The positive temperature rise of the composite insulator is significantly reduced after the adjustment.

II. OPERATING ABNORMAL HEAT OF COMPOSITE INSULATORS AND INFRARED PHOTO ACQUISITION

Insulators abnormal heat was found in a 500kV line of Guangdong Power Grid Corporation in June 2018. Two composite insulators were tested from the N133 and N49 towers of the 500kV line for, numbered 18C1 and 18C2. The above two composite insulators are made of two mainstream manufacturers in China, which were produced in May and September 2010 respectively, and were installed in December 2011. The typical heat generation and the falling of the corona ring are shown in Fig. 1.

The overall infrared characteristics of the insulators A and B were observed. It can be found that the high voltage end and the region near the high voltage end of the insulator B appeared obvious temperature rise, while the insulator A's temperature rise was mainly near the high voltage end, which indicates that the temperature rise is caused mainly by the corona discharge.

This abnormal heat is caused by the falling of the corona ring. Usually, the two ends of the insulator or the insulator string are respectively at the conductor's voltage potential and the ground voltage potential. The voltage is not uniformly forcing on the entire insulator. The distribution of the voltage on the insulator is that the voltage drop near the wire end is the largest, and the voltage drop far from the wire end is gradually reduced. When it is close to the cross arm of the tower, the voltage drop increases. Therefore, corona may occur at the high voltage side on the insulator. The principle of the corona ring (voltage equalizing ring) is to install a metal ring on the wire end of the insulator, which is equivalent to adding a parallel capacitor to increase the capacitance of the insulator to the wire and improve the

voltage distribution of the whole insulator. It can found in Fig. 1 that the corona ring of insulator B was fall on the fittings and distorted the distribution of the electric field around. In Fig. 1 the essential image captured by the drones from high altitude angle precisely described the abnormal heat problem of the insulators, which cannot be captured from the ground, proved the UAV promisingly of great feasibility to help the maintenance department to recognize potential fault.

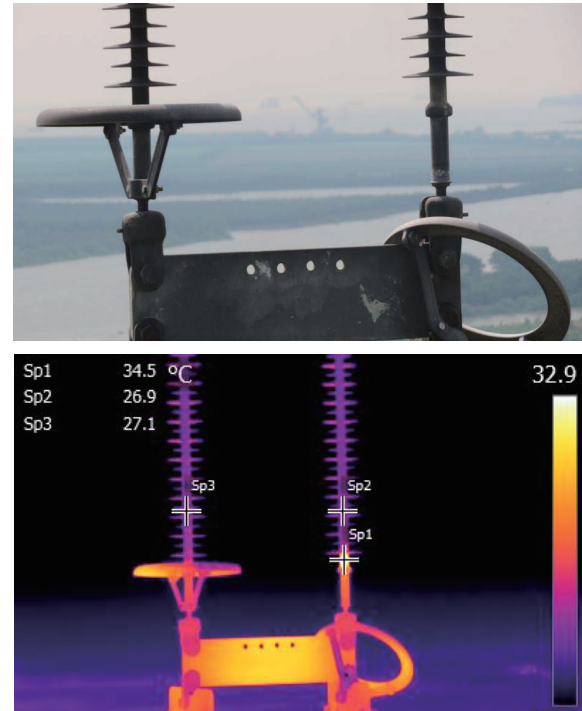


Fig. 1. Scene of the abnormal heat composite insulator on a 500kV line(left: insulator A, right: insulator B)

In Fig. 2 the key elements and operation mechanism of UAV patrol-inspection system to capture infrared picture of the insulators is displayed.

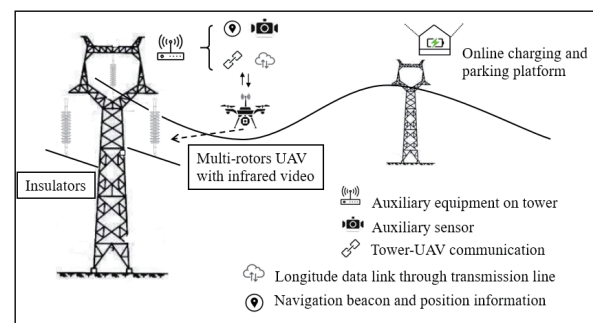


Fig. 2. Key elements and operation mechanism of UAV patrol-inspection system to capture infrared picture of the insulators

The drone is used to capture the infrared pictures of the composite insulators continuously and identify any suspicious heat and to share the information to the maintenance department to adopt further reaction.

Drones are the mobile camera that connect the observed objects to the data processing center. Different from the artificial picturing method from the ground, the advantage to capture images from the drones can be concluded to the shooting position which is close to the insulators and free from visual obstacles. The ground background can be

selected according to the actual situation to eliminate the heat interference of the background. Drones can respond quickly to infrared picturing tasks. The low cost has enabled users to expand the inspection range through the transmission system.

The auxiliary equipment on transmission tower shown in Fig. 2 integrates tower-drone communication, longitude data link through lines, and navigation beacon. Further, the insulator itself may also contain an electronic ID to identify and mark the image data. Through an effective information connection the infrared image is transmitted to the data processing center through the vertical communication of the power line, indicating that the drone can be a good substitute for the work of the ground workers.

To realize the comparison and evaluation of the historical data of the composite insulator based on image recognition, the first task is to capture the composite insulator image of high-quality. The insulator image captured in the line environment has a complex image background and the image clarity is affected. The limitation of shooting conditions, how to shoot, extract and clear the image of insulators is the core technic of the task, as well as the basic work of carrying out following tasks.

By extracting the edge of the insulator in the image, the insulator image is distinguished. It is also possible to determine a certain threshold by using the distinct difference in the gray value of the pixel at the boundary between the target insulator and the background image. However, it is difficult to identify images with relatively complex backgrounds. In the image captured by drones, the edge of the insulator has special outlines, which is distinctive from other devices on the power transmission line. Therefore, According to the repeating-outline characteristics of the insulator, the insulator can be identified based on the edge direction histogram method.

At present, the edge direction histogram (EHD) method has been widely applied to the extraction of image features. According to the block edge modes EHD can be divided into six types, including: horizontal edges, vertical edges, 45° edges, 135° edges, non-directional edges, and no edges. The top 5 edge modes are extracted to construct local and global edge histogram descriptors, which have a good description of the texture features of the image and effectively extract the target.

There are three common problems with currently photographed insulator images. In color images, a large amount of color information will take up more calculation time and affect the post-processing speed, which is the first problem to be solved by pre-processing. Since the image is too dark or too bright due to the illumination and the acquisition instrument during the acquisition process, the image contrast is not strong and the image features are not obvious. Finally, the noise and signals in the image are intertwined. Particle noise generated by exposure or photo electricity and random noise generated when digitizing images causes image quality to deteriorate. The inevitable interference generated during transmission and image generation degrades the quality of the image, too. So denoising is a very important part of image preprocessing. In the color image, the insulator should be a prominent part and the noise and environmental background should be the part of the suppression.

III. UAV INTELLIGENT INSPECTION COMBINED WITH MANUAL INTERVENTION TO PREVENT ABNORMAL HEAT

Drones integrates the advantages to improve the information fusion of time and space. The label information of the insulators and the infrared images are simultaneously transmitted and recorded.

With current technologies, it is still unrealistic to use the UAV to completely replace the manpower in the transmission system. But in some conditions, UAVs have shown unique advantages. However, it can be an ideal mode to use the UAV for intelligent line inspection which is combined with data center fault identification. The inspection system sends fault information and maintenance instructions to the engineering maintenance department to realize the fault prevention.

Applying UAV to the power line inspection and the fault prevention system of the power system can make full use of its superiority in cost and efficiency, which greatly solves various problems existed in the current manpower-maintenance mode [16]. Drones can carry visible cameras, infrared, ultraviolet cameras, ultrasonic and radar equipment, and electromagnetic field measurement equipment [17]. In this paper the infrared image is taken use to recognize the hidden faults.

The data transmit from the drones mainly includes infrared images data and fight data. Fig. 3 shows the flow of the main data in the ideal UAV intelligent inspection system. It can be seen that the drones and the communication equipment on the tower only need to temporarily store a small amount of data. The infrared data is sent to the data processing center for storage and analysis through the link of multiple tower relays. The position and time information of the drones will be used to locate the fault insulators and to record the weather information. From this chart the UAV can be concluded as a volant worker who is patrolling the transmission line safely and effectively.

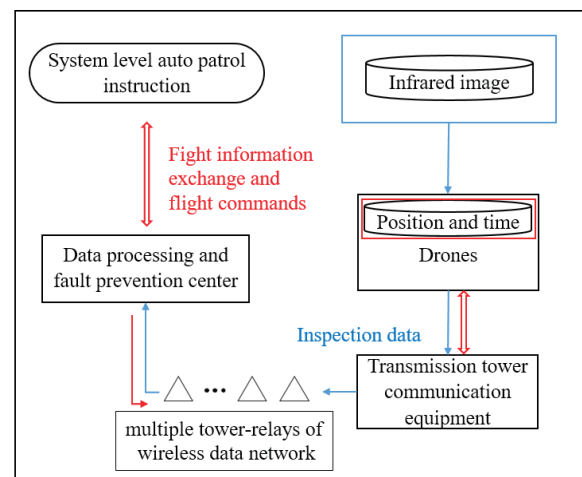


Fig. 3. Infrared images data and position beacon data flow among drones, tower equipments and the data center

The infrared image obtained by the drone can not only quickly judge the abnormal heat insulator, but also provide a reference for manual online maintenance. In Fig. 3, abnormal heat was determined by the data processing center based on the infrared image acquired by drones. After confirming the hidden fault that may cause more severe faults, the data

processing center will inform the maintenance department to intervene to remove the fault.

According to the operating defects founded by UAVs, the electric field distribution of the composite insulators at high voltage end was simulated with different voltage corona rings' shielding depth. The model with different shielding depth was shown as Fig. 4.

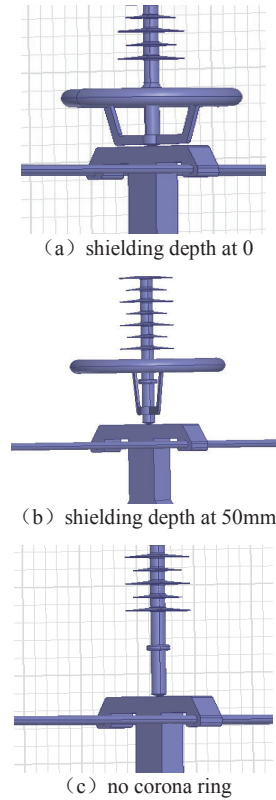


Fig. 4 Calculation model of different corona ring configurations for a composite insulator of a 500kV line

TABLE I. PEAK ELECTRIC FIELD STRENGTH PEAK RESULTS OF COMPOSITE INSULATORS WITH DIFFERENT CORONA RINGS CONFIGURATIONS

Corona Ring State	Maximum Electric field strength (kV/cm)
Shielding depth at 0mm	6.3
Shielding depth at 50mm	4.3
No corona ring on high voltage end	16.8

The maximum electric field strength at the high voltage end is shown in Table I. It can be seen that the electric field strength of the high voltage end of the sample without the corona ring is the largest, being 16.8 kV/cm; the shielding depth remains 0 mm of the corona ring, the electric field strength is 6.3 kV/cm. When the shielding depth reaches 50mm, the electric field strength is the smallest (4.3kV/cm). It is informed that by adjusting the shielding depth to 50mm the electric field distortion can be effectively reduced, which can avoid the heating defects.

It is believed that after the composite insulator is accumulated with pollution, the polarization loss generates heat under the action of a strong electric field at the high voltage end. At the same time, due to the insufficient shielding depth of the corona ring which may be caused by

the falling of the corona ring, the field distortion of the high voltage end is further increased. This leads to the severe development of the polarization loss, and causes different degrees of heating at the high voltage end of the operating composite insulator.

In fig. 4 The corona ring bracket is added by hot-line working to increase the shielding depth. Infrared temperature measurement is performed on the adjusted end of the insulator. It can be seen in Table II that after the equalization ring adjusted, the abnormal heat generation at the high-voltage end of the composite insulator drops significantly. The temperature rise is reduced from 5.6K and 4.8K to 0.6K and 0.9K after adjustment, indicating the effective fault-prevention function of the proposed mechanism.

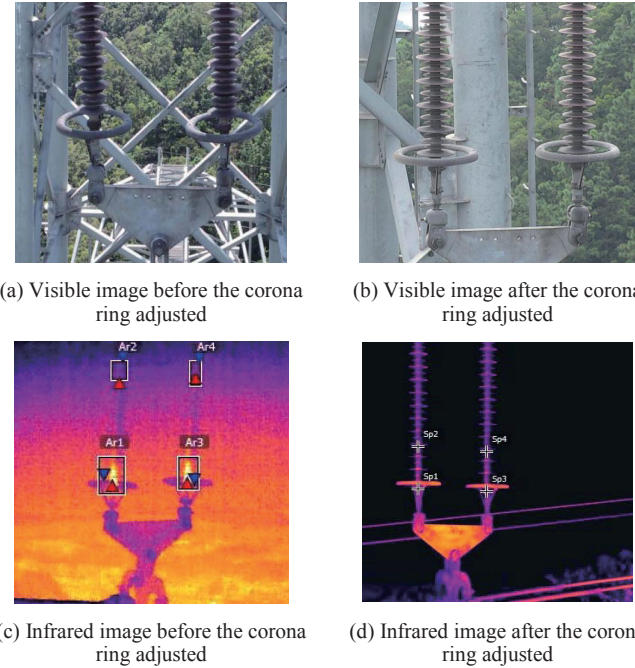


Fig. 4. Infrared and visible image of the corona ring adjustment for a 500kV line composite insulator

TABLE II. INFRARED TEMPERATURE MEASUREMENT RESULTS OF A 500KV LINE COMPOSITE INSULATOR EQUALIZING RING

Composite insulator sample	Corona ring adjustment	Temperature rise (K)
N39 tower A phase left	before	5.6
N39 tower A phase right side		4.8
N39 tower A phase left	after	0.6
N39 tower A phase right side		0.9

Through the test and analysis of a composite insulator with abnormal heat of a 500kV line, it is found that the heat part is only the surface layer of the sheath, showing typical surface heat phenomenon. The heat mechanism is the polarization loss heat under the action of strong electric field.

From the UAV intelligent inspection finding and recognizing the potential fault of the insulator's abnormal heat, to the data center to identify the fault, and then to the early warning maintenance department to send manpower to solve the fault, the fault monitoring and prevention system is fully simulated in this paper. Although the fault intelligent identification part and remote communication part are still to be built, this mechanism has been clearly proposed and can

be a key construction area in the development of smart grid transmission networks in the future.

IV. DISCUSSION

In this paper, a UAV intelligent inspection system is proposed to monitor and prevent abnormal heat of composite insulators of 500kV transmission system. The reason of the insulators' abnormal heat is well discussed based on infrared images captured by the UAV. By adjusting the position of the shielding rings of operating composite insulators, the temperature rise of the composite insulator decreased significantly. The mechanism how the UAV intelligent inspection system operates and the key technology to realize fault monitoring and prevention is well discussed. It is proved that the UAV is promisingly of great feasibility to help the maintenance department to recognize potential fault on the transmission lines and prevent severe faults.

For now, the identification of faults relies on professionals to observe and compare the visible and infrared images. This method is inefficient and requires a lot of manual work. To apply drones to realize the automatic identification and early warning of insulator faults, on the one hand, it requires the communication and data center construction throughout the whole grid. On the other hand, building the expert fault identification system is also important to improve the intelligence of the fault monitoring and prevention system. Combined with manual marking and simulation study, how to learning faults from infrared images and varieties of other data is the promising direction that deserves further study.

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